Kolade Alabi

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Summary

ML Engineer & Graduate Researcher at University of Maryland, College Park with 3+ years experience shipping production-grade ML features and resilient backend systems. Currently focused on developing theoretical frameworks for robust optimization in ML systems to make AI safer.

Experience

University of Maryland, College Park
College Park, MD

Graduate Research Assistant — *January 2026 - May 2026*

- Developed theoretical co-training Machine Learning frameworks for reliable and high-utility Human-AI collaborative recommendations
- Investigated robust optimization techniques for minimizing sample complexity and training in practical settings

JPMorgan Chase
Houston, TX

Software Engineer — *August 2023 - August 2025*

- Spearheaded the design and development of an Agentic LLM Assistant for presentation creation, fit with Q&A support capabilities as well as integrations with newly created VectorDB internal knowledge base for RAG, content library APIs, and internal financial applications, each equipped with safety filters to ensure reliable and firm-compliant responses
- Developed the core platform for the AI Assistant, a .NET PowerPoint add-in that streamlined business presentation creation for over 80,000 users
- Modernized core project APIs to AWS hosting, including the refactoring of Apache Solr-powered search to AWS OpenSearch full-text querying — increasing scalability and allowing for retirement of global physical servers
- Execution Excellence Award Q1 2024: One of 5 recipients out of ∼200 JPMC SEP engineers for contributions to the modernization of key application modules and sharing of expertise on AWS ECS and OpenSearch

Carnegie Mellon University
Remote

Research Intern (Part Time) — *July 2023 - December 2024*

- Created Singularity container to run the Deep Iterative Subtomogram Clustering Approach (DISCA) deep learning pipeline end-to-end and furthermore trained CMU collaborators from multiple universities on the usage of said container, reducing setup time by nearly 80%

Capital One
McLean, VA

Software Engineering Intern — *June 2023 - August 2023*

- Engineered and deployed Universal Deep Link servicing — powering seamless integration for partners like Walmart, Google Autofill, and Zelle — driving increased traffic to the EASE mobile application while reducing navigation time by up to 75% for ∼38 million existing EASE users
- Deployed a PostgreSQL database on Amazon Aurora, eliminating the use of a middleman service and reducing expenses by 19%

Education

University of Maryland, College Park
College Park, MD

Master of Science in Computer Science, GPA: 3.8/4.0 — *August 2025 - est. December 2026*

- **Key Coursework**: Theory of Robust Machine Learning, Generative AI Agents, Language Model Interpretability, Systems for Machine Learning, Uncertainty Communication for Decision-Making, Interactive Technologies in Human-Computer Interaction, Advanced Computer Graphics

The University of Texas Rio Grande Valley
Edinburg, TX

Bachelor of Science in Computer Science, GPA: 3.8/4.0 — *August 2022 - May 2023*

Google
Remote

Computer Science Research Mentorship Program — *February 2023 - May 2023*

- Collaborated with Google DeepMind NLP researcher to explore best practices in academic Machine Learning research
- Invited to present research findings on novel applications of Transfer Learning architectures to predict the progression of Alzheimer's in patients by way of blood samples to CSRMP participants

Projects

By Its Cover | PyTorch, ONNX, CLIP, LanceDB, Python, TypeScript, React, .NET 10, Docker, AWS (Lambda, S3, SQS, Batch)

- Architected a multi-vector embedding-based search engine to support for vague natural-text searches for books by way of a late-Interaction approach, enabling fine-grained spatial queries for both cover images and book metadata
- Developed a CLIP embedding-based hybrid Collaborative Filtering model for recommending book covers to users, taking advantage of features from both user relationships (by ratings) as well as aesthetic qualities from cover images
- Implemented an automated database ingestion pipeline, fit with book detail sourcing from both GoodReads and the HardCover API, to populate a LanceDB vector database of both book cover CLIP and two-tower network embeddings, enabling semantic search and recommendation capabilities
- Deployed a serverless book cover embedding endpoint to AWS Lambda, utilizing ONNX to optimize and serve the CLIP model asynchronously, reducing cold-start overhead and inference times by more than 90%

Skills

Machine Learning: PyTorch, TensorFlow, ONNX, Scikit-learn, SciPy, LangChain/LangGraph, Vowpal Wabbit, Pandas/Polars, NumPy

Databases: LanceDB, DuckDB, SQLite, MySQL, PostgreSQL, AWS RDS, MongoDB, OpenSearch/ElasticSearch

Languages/Frameworks: Python, FastAPI, C#, .NET, JavaScript/TypeScript, React, C/C++, R, Git, Terraform, AWS, Docker, GitHub Actions